

DTypeWithConstraints#

https://pytorch.org/docs/stable/generated/torch.quantization.backend_conf

Description:

Config for specifying additional constraints for a given dtype, such as quantization value ranges, scale value ranges, and fixed quantization params, to be used in `DTypeConfig`. The constraints currently supported are: `quant_min_lower_bound` and `quant_max_upper_bound`: Lower and upper bounds for the minimum and maximum quantized values respectively. If the `QConfig`'s `quant_min` and `quant_max` fall outside this range, then the `QConfig` will be ignored. `scale_min_lower_bound` and `scale_max_upper_bound`: Lower and upper bounds for the minimum and maximum scale values respectively. If the `QConfig`'s minimum scale value (currently exposed as `eps`) falls below the lower bound, then the `QConfig` will be ignored. Note that the upper bound is currently not enforced. `scale_exact_match` and `zero_point_exact_match`: Exact match requirements for scale and zero point, to be used for operators with fixed quantization parameters such as `sigmoid` and `tanh`. If the observer specified in the `QConfig` is neither `FixedQParamsObserver` nor `FixedQParamsFakeQuantize`, or if the quantization parameters don't match, then the `QConfig` will be ignored.

Function Signature

```
class
torch.ao.quantization.backend_config.DTypeWithConstraints(dt
type=None, quant_min_lower_bound=None,
quant_max_upper_bound=None, scale_min_lower_bound=None,
scale_max_upper_bound=None, scale_exact_match=None,
zero_point_exact_match=None)[source]#
```

Detailed Documentation

Documentation

Config for specifying additional constraints for a given dtype, such as quantization value ranges, scale value ranges, and fixed quantization params, to be used in `DTypeConfig`.

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